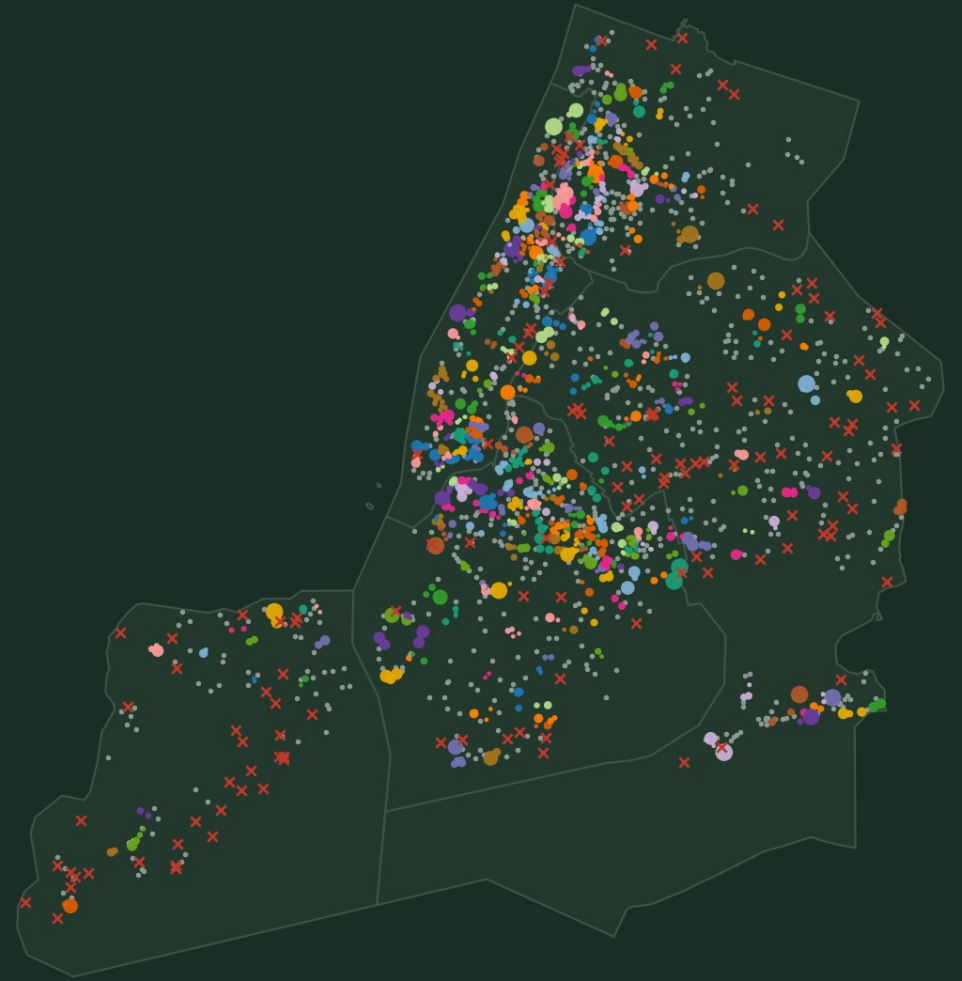


How Are New York's Parks Connected by Streets?

Grouping 1,902 small and medium parks by the shape of the walking network — not just by distance

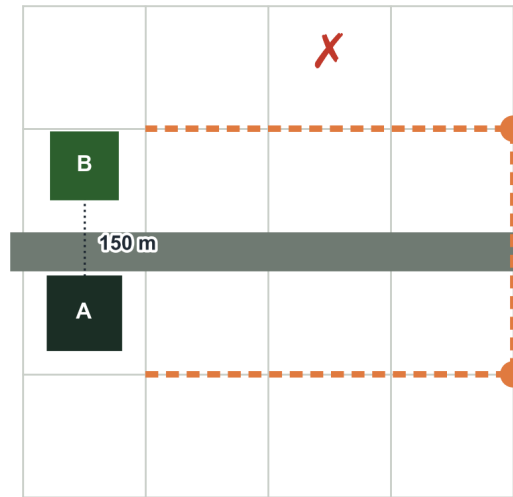
NYC Parks properties · OpenStreetMap walking network · Cityseer 5.8
October 2026



Close is not the same as connected

Why: next, we test whether connected groups of parks go with more shops and visits nearby.

- We usually ask how far apart two parks are. But people pick routes that are easy to follow.
- A route with few turns feels simple — like walking along one straight avenue.
- So we ask: which parks are linked by the street layout itself?



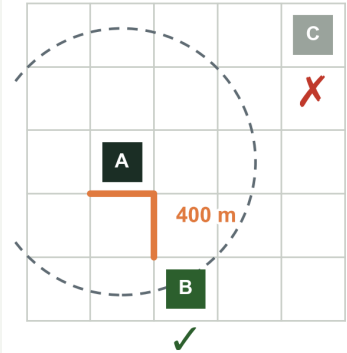
Close on the map — but you must walk around a barrier



Farther apart — but one straight street

This idea comes from Space Syntax, a field that studies how the shape of streets guides where people walk.

Four ways two parks can be “connected”

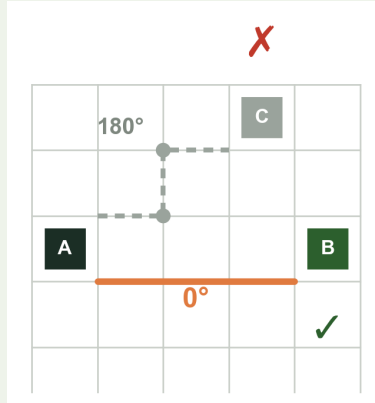


D

Short walk

The two parks are only a few minutes' walk apart along real streets (200–800 m).

The usual “nearby” — our baseline.

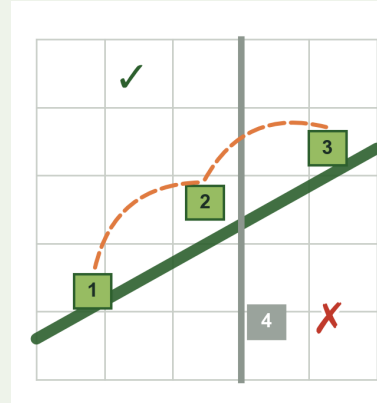


N1

Few turns

You can walk from one park to the other while turning very little.

Directions that say: “just go straight.”

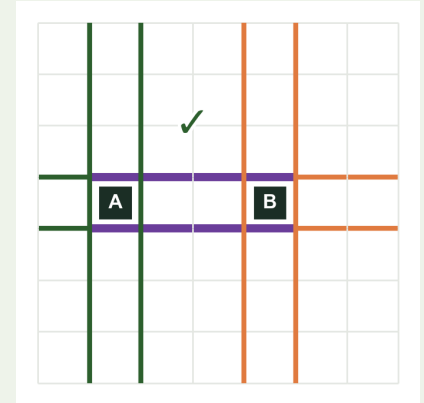


N2

Same street axis

Both parks sit along one continuous street you can follow without switching streets.

Two shops on the same avenue.



N3

Shared walking zone

The streets reached by walking (almost) straight from each park overlap more than distance explains.

Two schools whose students use the same streets.

For all four: every park in a group must be within a 10-minute walk (800 m) of every other park in that group.

What we used

1,902

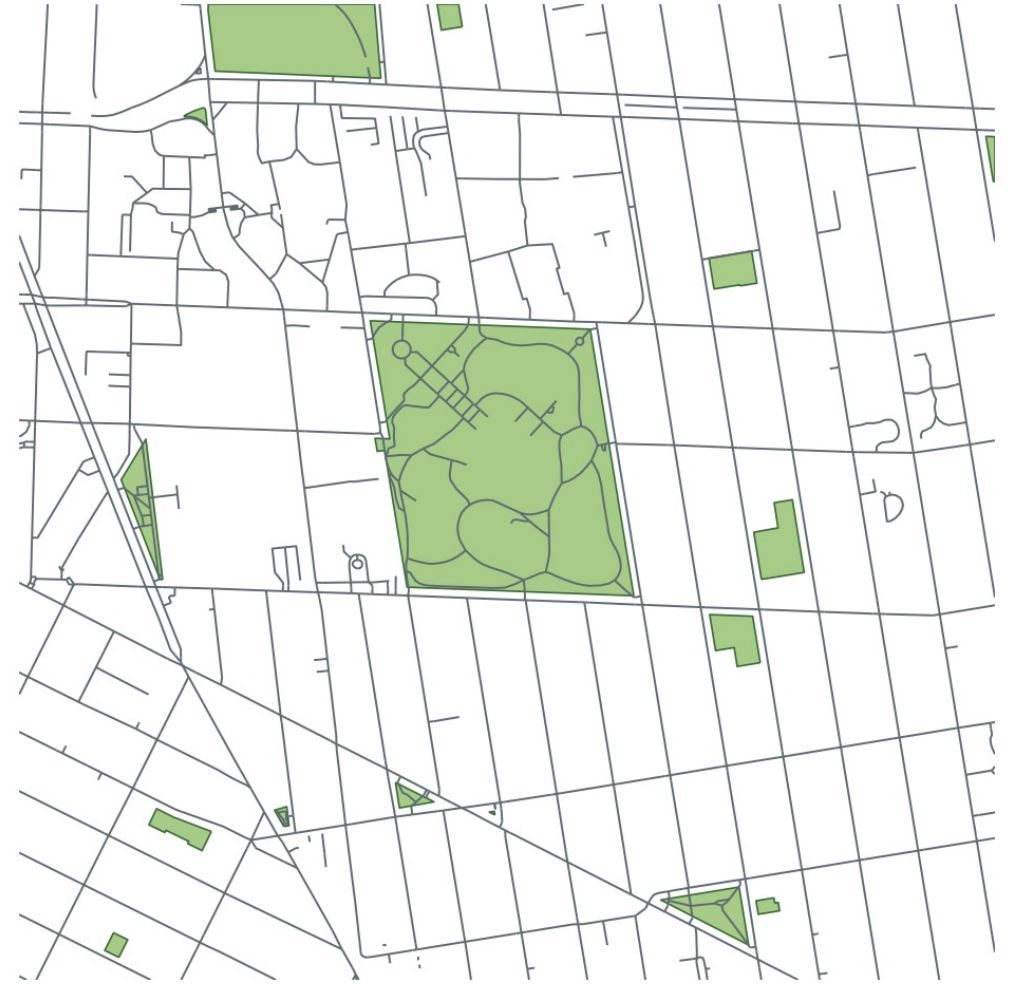
small & medium parks — we left out huge parks such as Central Park, and lots that are not real parks

496k

walkable streets and paths from OpenStreetMap, the free “Wikipedia of maps”

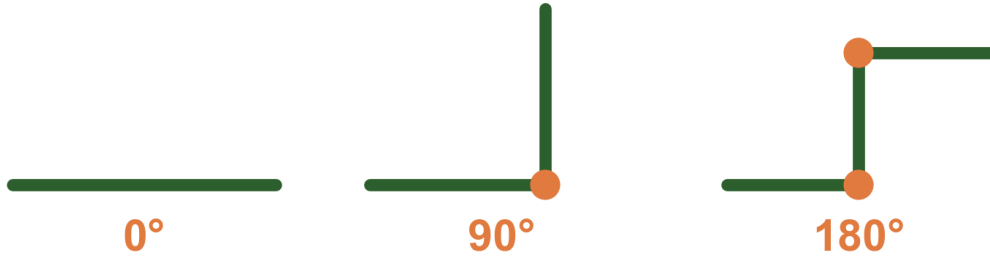
25 → 180

very long parks (e.g. parkway medians) cut into ~400 m pieces, so one park cannot link places kilometres apart



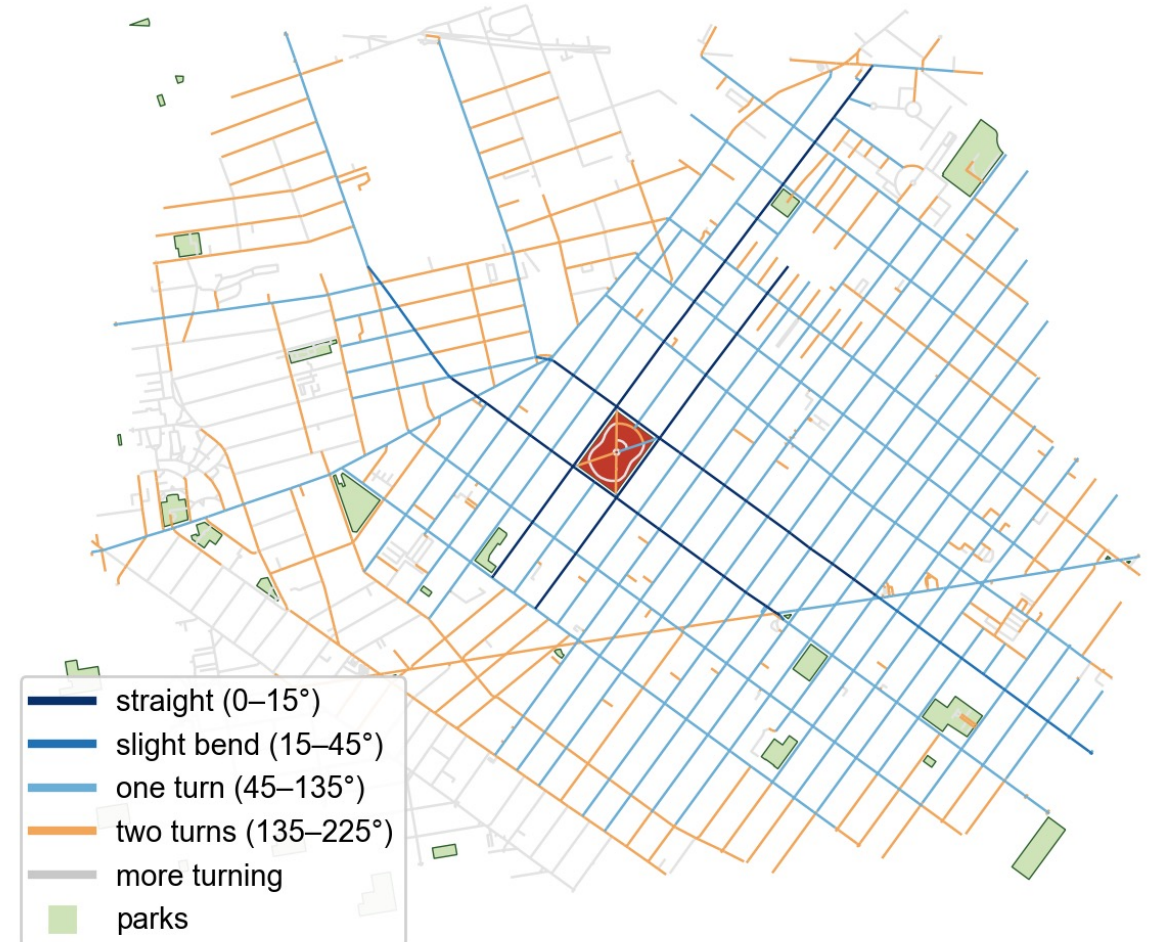
Walking network (lines) and parks (green) around Fort Greene, Brooklyn

Counting how much a route turns

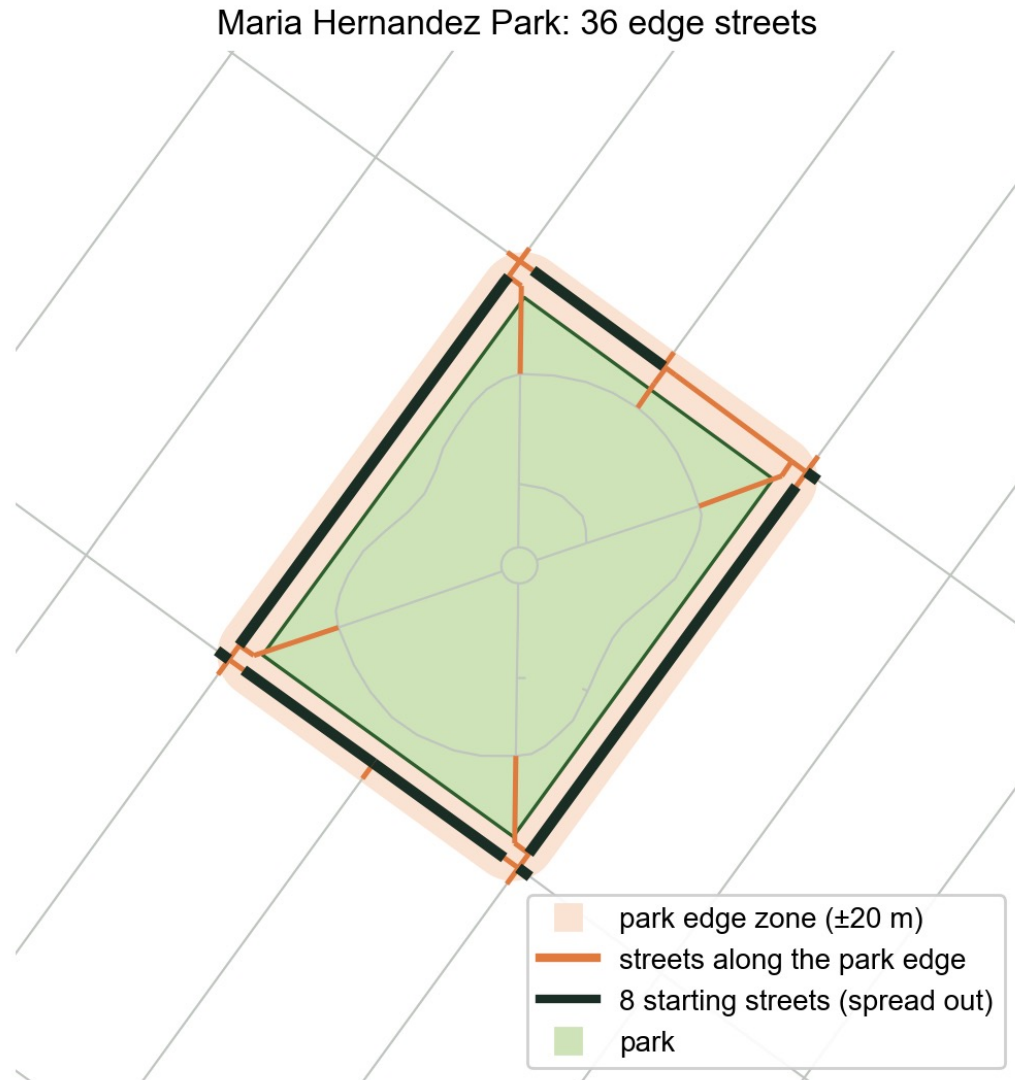


- Going straight = 0° . Turning a corner $\approx 90^\circ$.
- Between two parks we find the route that turns the least (not the shortest one) and add up its turns.
- Map: from one park, dark-blue streets are reached by going straight, light blue with one turn, orange with two.
- Computed with Cityseer, an open-source tool for analysing street networks.

Turning needed to reach each street from Maria Hernandez Park



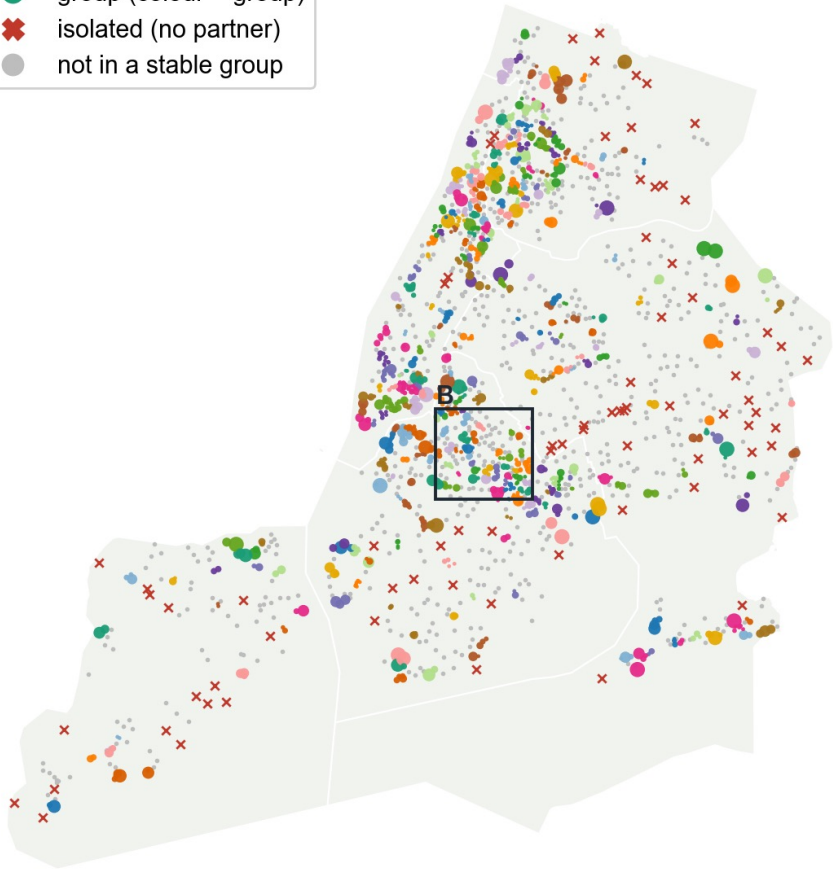
Where does a park meet the street?



- We don't know where each park's entrances are.
- So we treat the streets running along the park's edges (orange) as its doors.
- Big parks have many edge streets, so we start from at most 8 spread-out ones (black). This keeps big parks from getting an unfair advantage.

D · Parks within a short walk (baseline)

- group (colour = group)
- ✕ isolated (no partner)
- not in a stable group



B · Fort Greene – Bed-Stuy – Crown Heights



349

groups

1,274

parks in a group

3

parks in a typical group

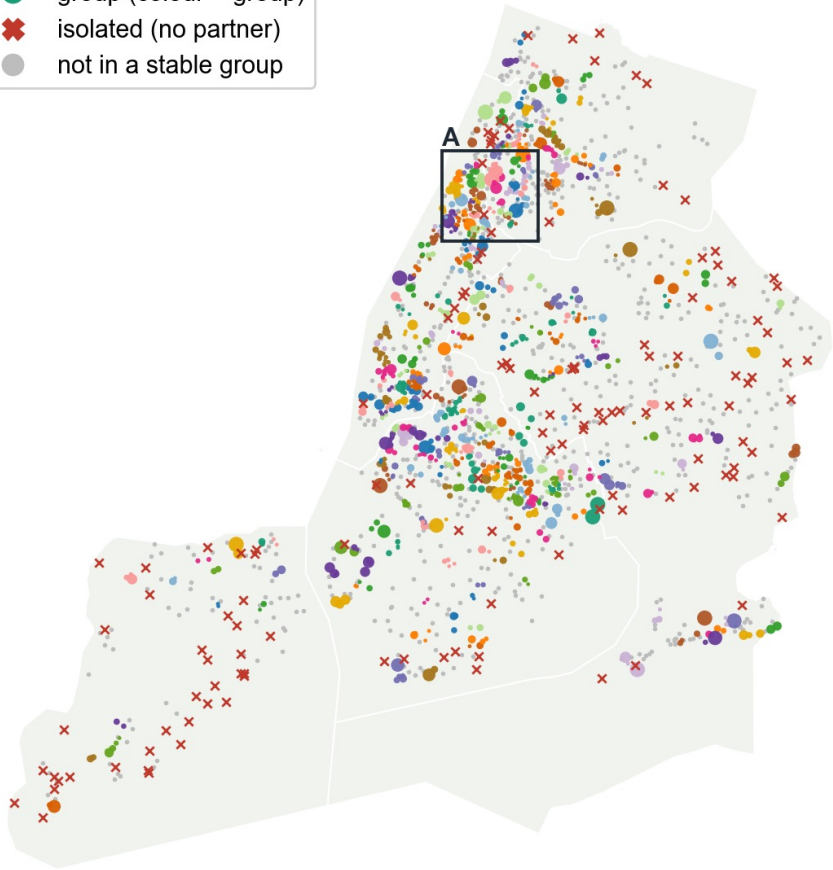
76

parks with no other park nearby

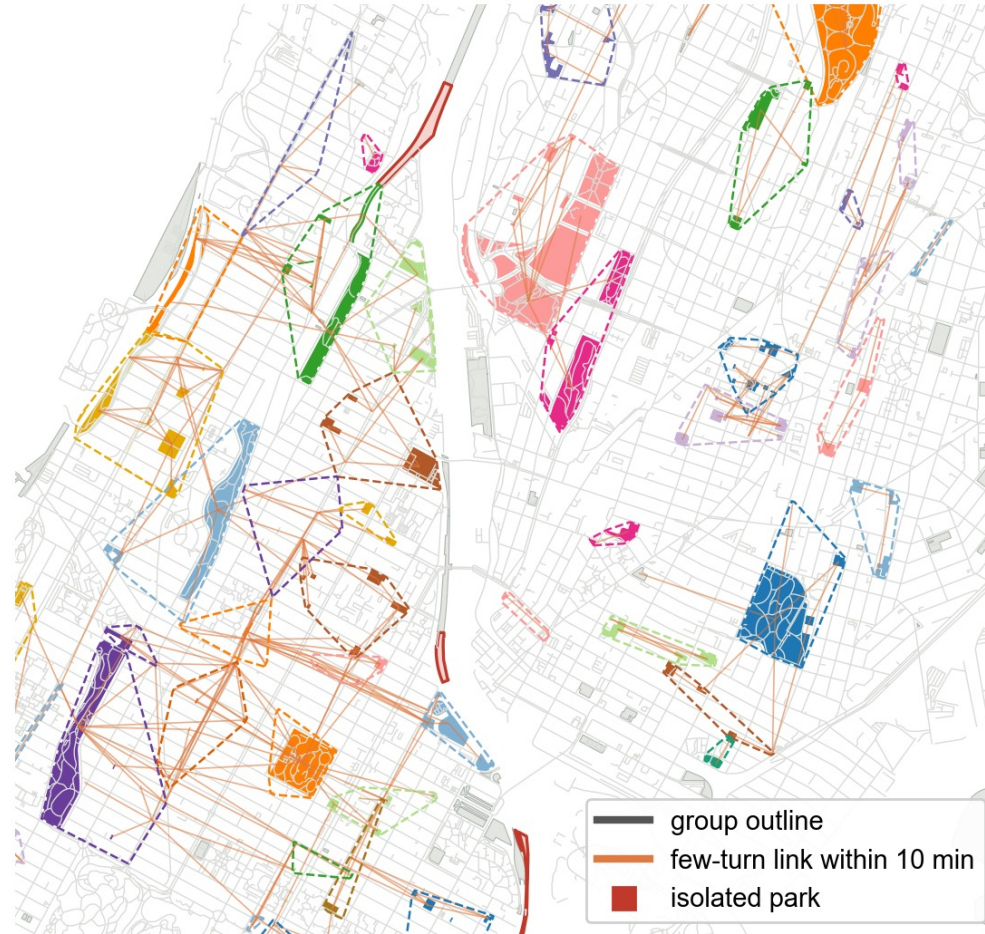
Each colour is one group. This is pure proximity — the other three definitions are compared against it.

N1 · Parks linked by few-turn routes

- group (colour = group)
- ✕ isolated (no partner)
- not in a stable group



A · Harlem / Upper Manhattan – South Bronx



340

walkable groups

1,178

parks in a group

3

parks in a typical group

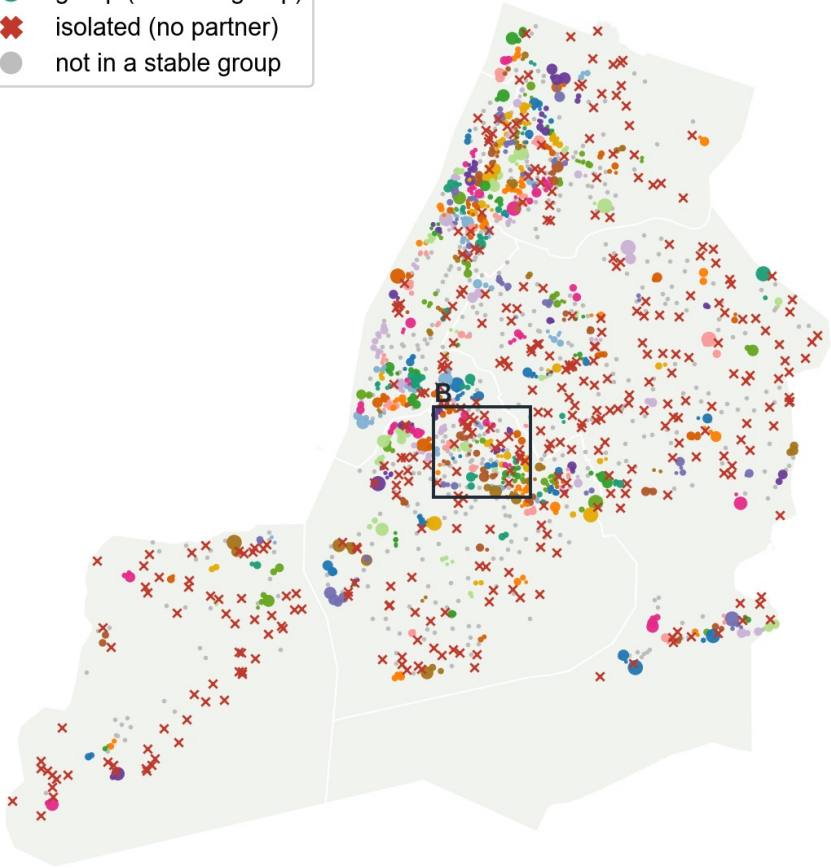
132

parks with no few-turn partner

Each colour is one group. Orange lines: few-turn links within a 10-minute walk.

N2 · Parks along the same street axis

- group (colour = group)
- ✕ isolated (no partner)
- not in a stable group



B · Fort Greene – Bed-Stuy – Crown Heights



329

groups

1,212

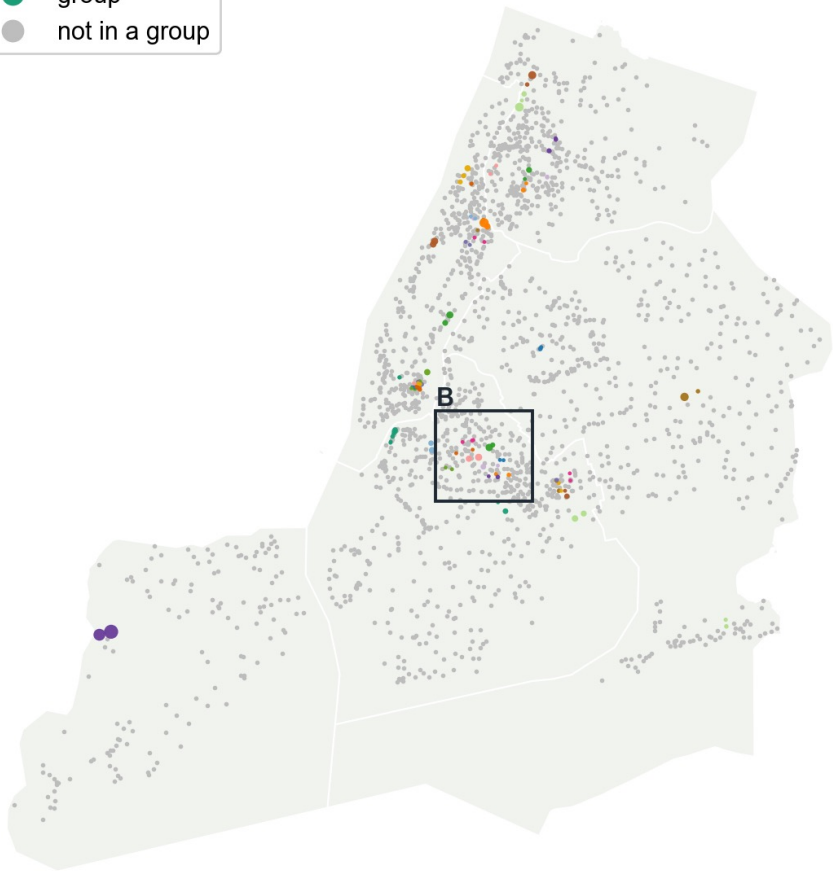
parks in a group

“Street axis” = one continuous street you can follow without switching streets (direction change $< 30^\circ$ at each junction).

46% of N2 groups touch a main street (top 5% of through-walking) — kept as a group attribute.

N3 · Parks sharing a walking zone

- group
- not in a group



B · Fort Greene – Bed-Stuy – Crown Heights



47

groups (mostly pairs)

113

parks in a group

Rare: two parks' straight-walk zones seldom overlap more than their distance alone explains.

N3 is a weak signal — useful mainly as a contrast to N1 and N2.

Groups must be walkable: the 10-minute rule

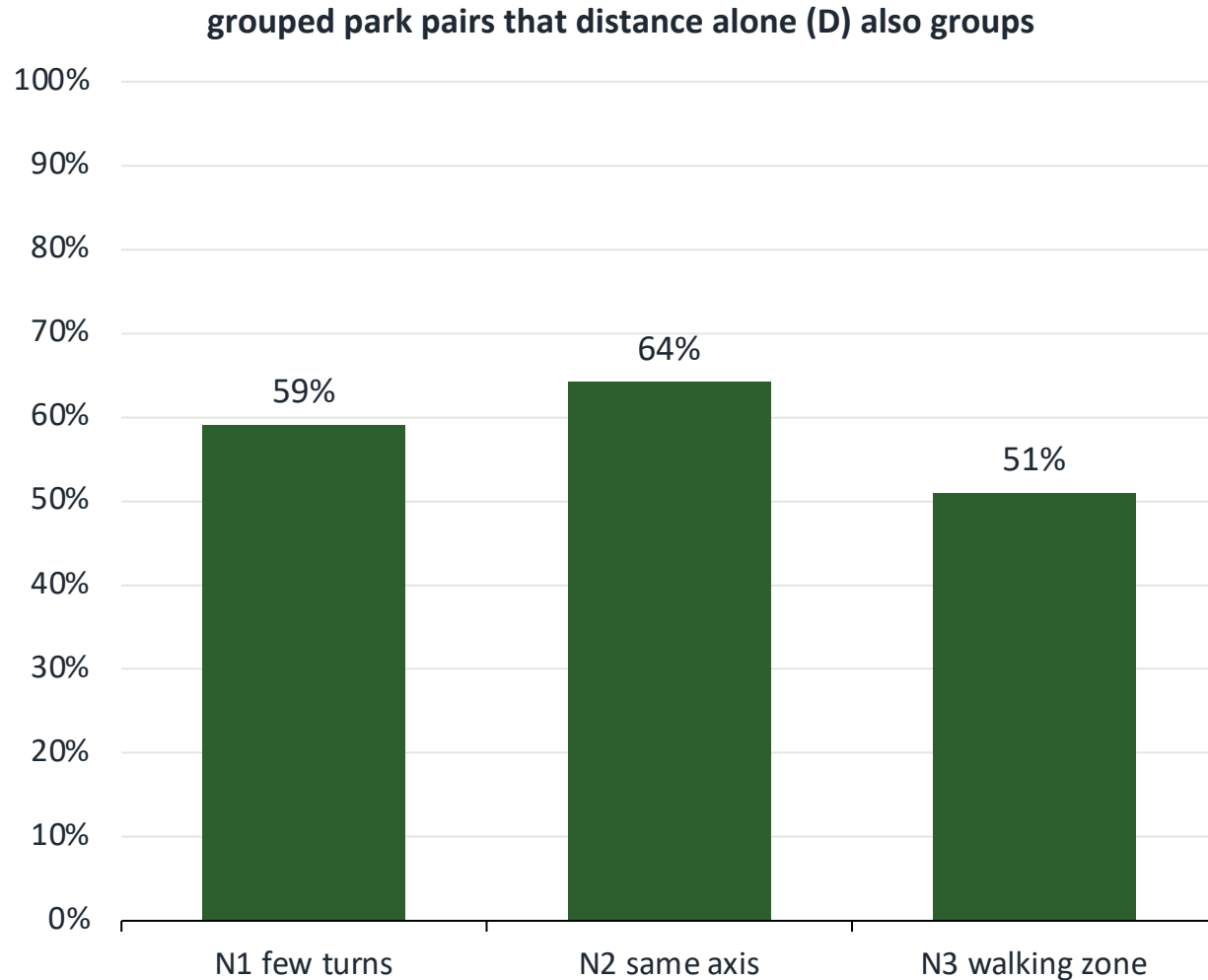
Brooklyn: 17 parks spread over 2.2 km on foot → 5 walkable groups



- Street-shape links can chain parks across several kilometres.
- So inside each group, every pair of parks must be within a 10-minute walk (800 m along real streets).
- Long groups are split; a park with nobody within 10 minutes leaves the group.

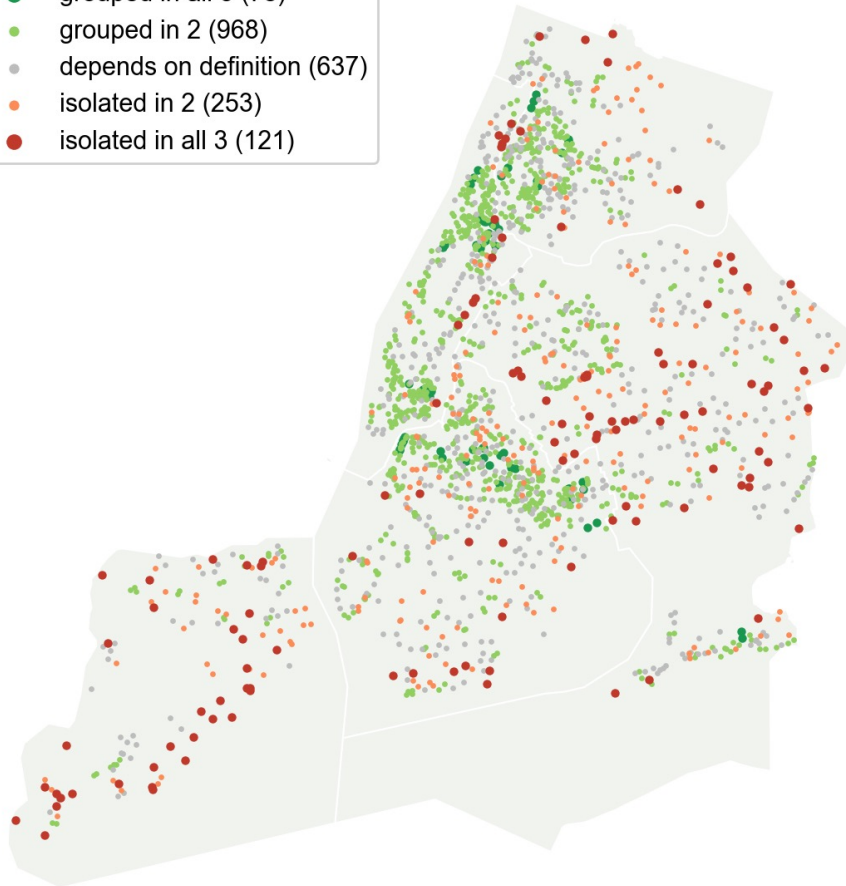
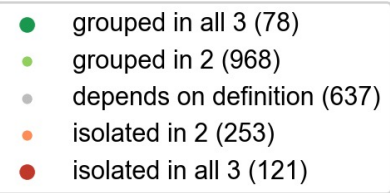
	groups	largest group
D short walk	281 → 349	29 → 16
N1 few turns	217 → 340	41 → 15
N2 same axis	231 → 329	32 → 22
N3 walking zone	63 → 47	8 → 8

Is this just about distance?

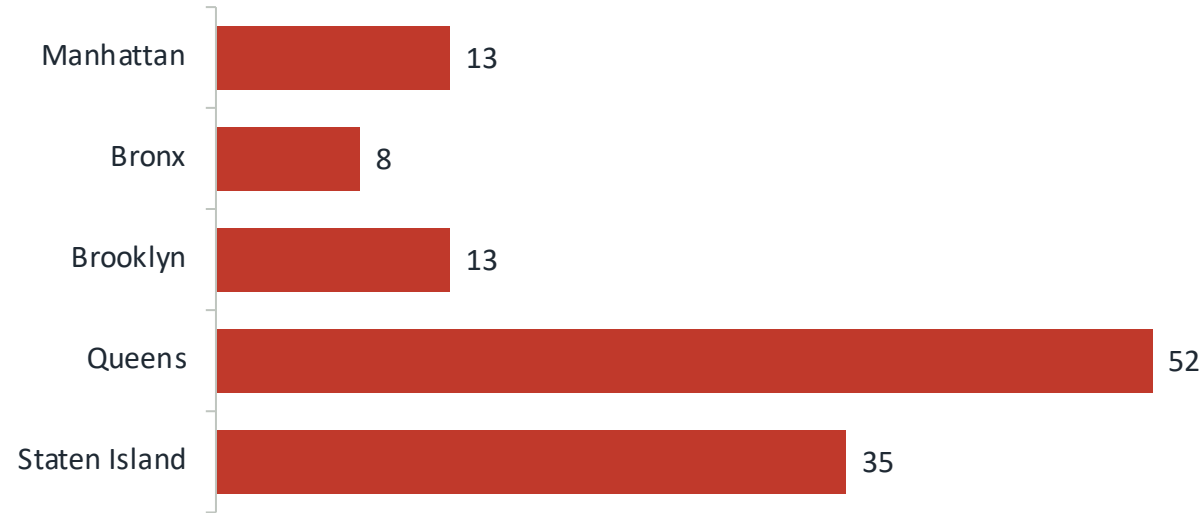


- We compare each definition with D, which groups parks by walking distance only.
- N1: 59% of its grouped pairs are also grouped by D — the other 41% come from street shape.
- N2: 64% · N3: 51%.
- So street shape changes which nearby parks belong together.

Parks left alone



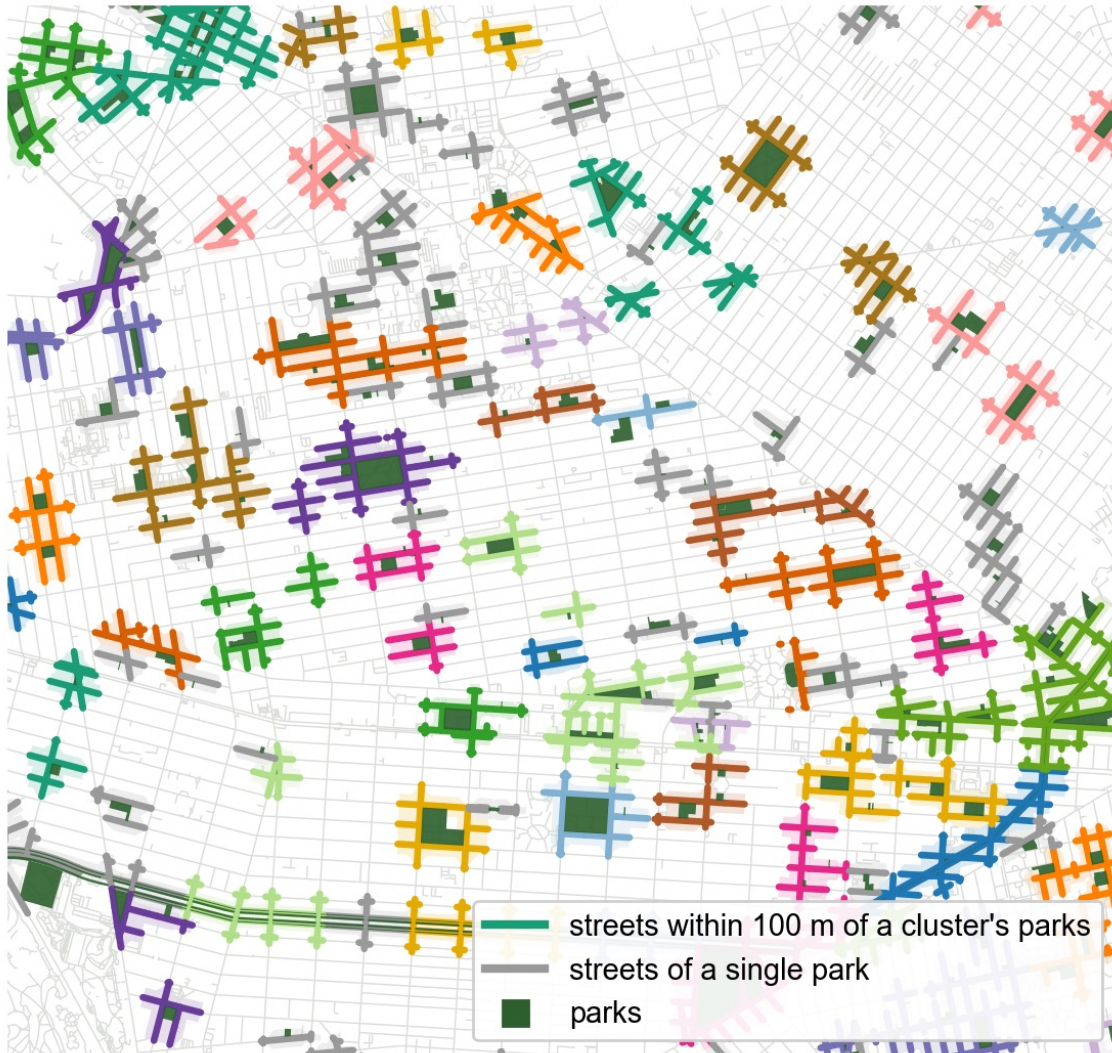
Isolated under all three definitions (121 parks)



- Mostly in Staten Island and Queens, where streets curve and dead-end.
- Also parks squeezed next to highways or along the waterfront.
- They sit outside the walkable “park network” — worth a look for planning.

Next step: parks and commercial activity

N1 clusters: streets within 100 m along the network (zoom B)



- Every park belongs to one cluster per definition (a single park counts as its own cluster).
- A cluster's commercial area = the streets within 100 m of its parks, measured along the walking network.
- A shop near two parks counts once, for the park that is closer on foot.
- Next: count shops, business variety and visits in each area (Advan POIs) and compare clusters across D, N1, N2 and N3, controlling for park size, population, jobs and street accessibility.

Ready now: park and cluster tables with street, population, income, jobs, land-use and subway controls.

Takeaways

Street shape matters

Few-turn links form 340 walkable park groups across NYC; 132 parks have no few-turn partner.

Not just distance

41% of few-turn group pairs are not grouped by walking distance alone.

Four lenses

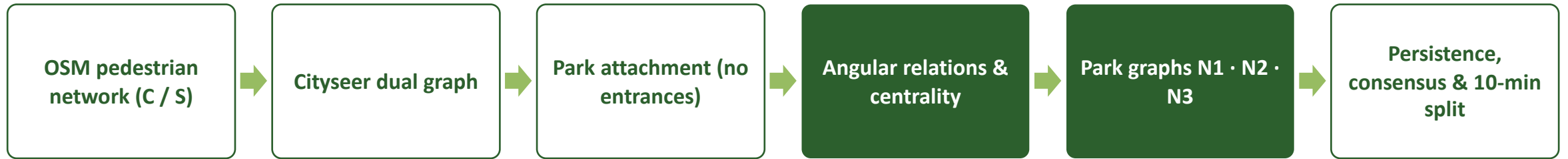
Distance, few turns, same street axis and shared zone give different groups — ready for the commercial analysis.

Next: measure shops and visits on the streets within 100 m of each cluster and run the regressions.

Appendix · Technical details

- Pipeline and tools
- Network representation and park attachment
- Angular cost on the dual graph
- D , $N1$, $N2$, $N3$: formal definitions and parameters
- From park graphs to groups (percolation)
- Proximity check
- Limitations

Pipeline and tools



- Parks: NYC Parks properties (2,061) → flagship parks, non-park parcels and parks > 100 acres removed → 1,902 parks; 25 long linear parks split into 180 pieces (2,057 graph nodes)
- Network: OSM via OSMnx with a custom walk filter (keeps streets tagged sidewalk=separate), NYC + 3.2 km buffer; borough networks with 3.2 km non-reporting buffers
- Software: Cityseer 5.8 (dual graph, simplest/shortest-path Dijkstra, angular centrality), NetworkX (graphs, Louvain), GeoPandas/Shapely

Network representation and park attachment

S · sidewalks & crossings drawn separately
2,054 segments



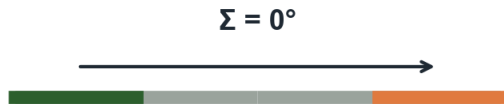
C · one line per street + footpaths (main)
876 segments



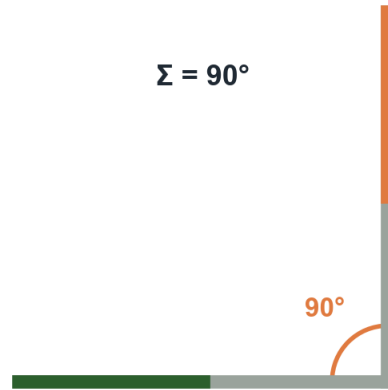
- S draws sidewalks and crossings separately: parks facing each other across a street are $\sim 180^\circ$ apart.
- C (street centrelines + pedestrian-only paths) behaves like a Space Syntax segment map $\rightarrow$ main analysis; S = sensitivity.
- Attachment: segments within ± 20 m of the boundary whose in-band length $\geq 50\%$ of the segment or $\geq \min(2.5 \cdot d, 25\% \text{ of perimeter})$
- K = 8 sources by farthest-point sampling; “unresolved” if attached only to components < 200 segments

Angular cost on the dual graph

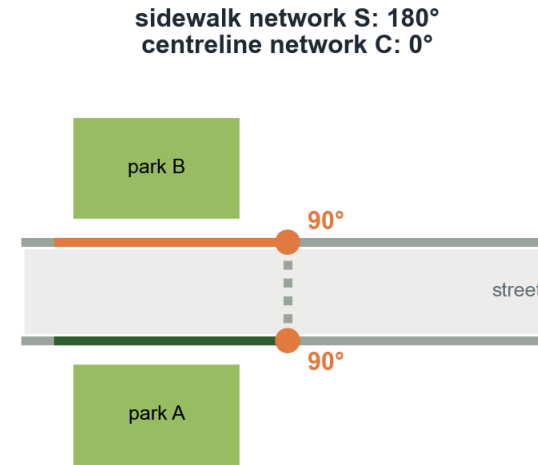
(a) Straight



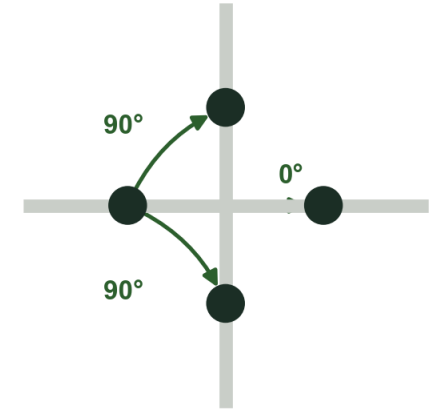
(b) One turn



(c) Crossing the street



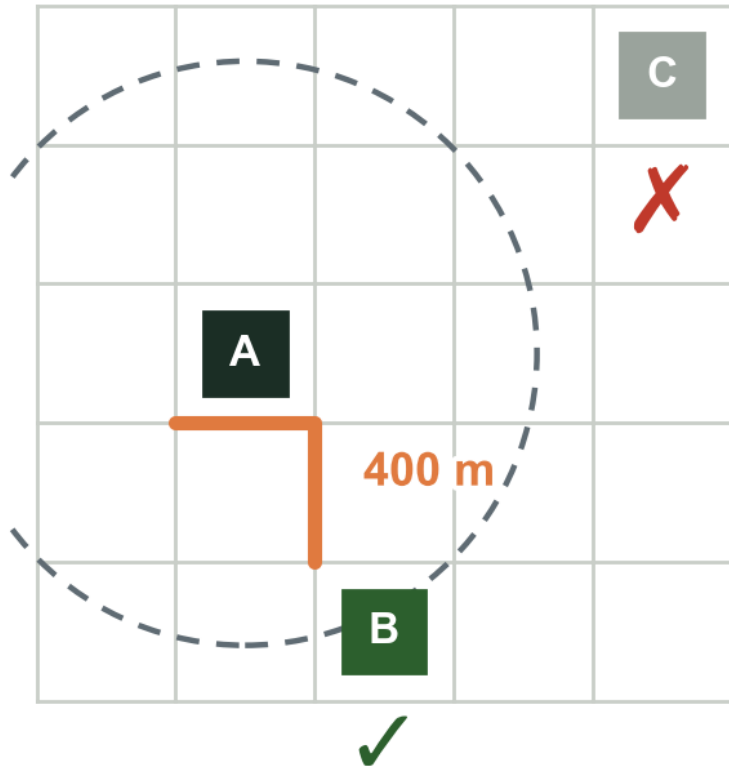
(d) Cityseer dual graph



Simplest path Cityseer routes on the dual graph (street segments = nodes) and minimises the cumulative change of direction in degrees, including bends inside segments.

Metric bound Each search runs up to 3.2 km; $R \in \{800, 1600, 3200\}$ m filters the path length afterwards — distance bounds the search, it does not define the edge.

D · Walking distance (formal)

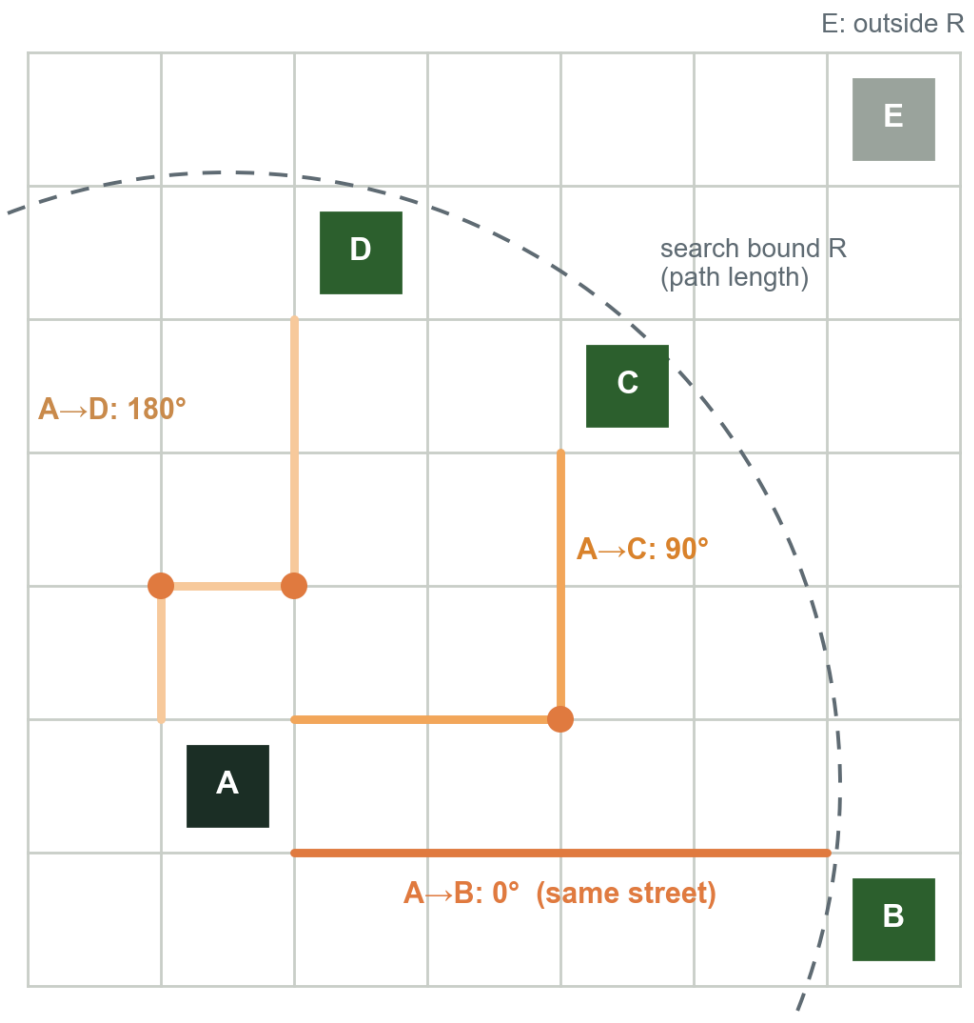


edge $\Leftrightarrow D_{ij} \leq d$

D = shortest walking distance between the two parks' attachment segments (Cityseer shortest-path trees)

- 3 settings: $d \in \{200, 400, 800\}$ m
- Includes adjacent parks (shared frontage, touching polygons) — proximity itself
- Same grouping procedure as N1–N3 (percolation filter, Louvain $\gamma = 2$, consensus, 10-min split): the proximity baseline for RQ3

N1 · Formal definition



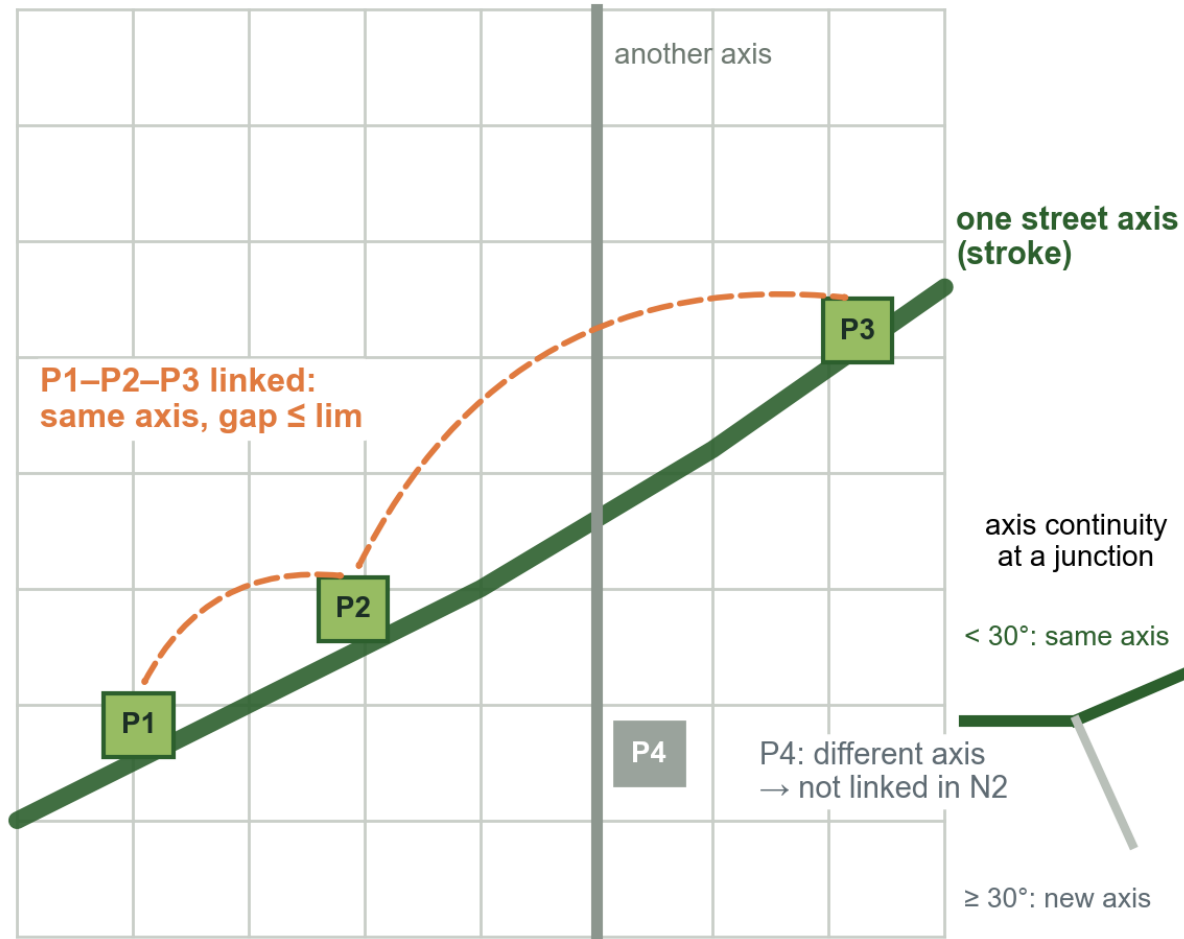
$A_{ij}(R) = \min \text{ over source } u \in B_i, \text{ target } v \in B_j \text{ of angular cost}(u \rightarrow v), \text{ path length} \leq R$

edge $\Leftrightarrow A_{ij} \leq \theta$

θ	reading	edges from A
15°	same straight street	B
45°	nearly straight	B
90°	one right-angle turn (strict)	B, C
180°	two turns	B, C, D

- 24 settings: $\theta \in \{15, 45, 90, 180\}^\circ \times R \in \{800, 1600, 3200\} \text{ m} \times \{\text{best pair}, \geq 2 \text{ robust pairs}\}$
- Groups: Louvain communities on the 9 non-percolating settings (weights $\exp(-A/45^\circ)$), consensus ≥ 0.5 , then split to a 10-min walking diameter
- Measured one-turn routes cost $\sim 90\text{--}100^\circ$ (bends inside segments), so $\theta = 90^\circ$ is strict

N2 · Street-axis continuity (formal)



edge $\Leftrightarrow$ same stroke and $\text{gap}_{\text{along}} \leq \text{lim}$

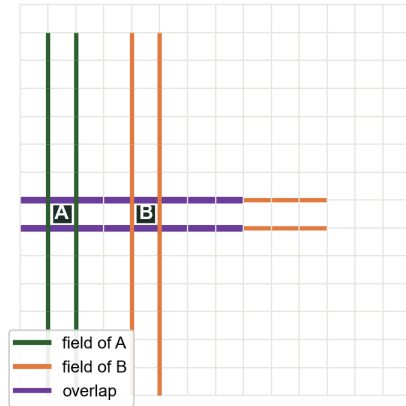
stroke = segments chained where the direction changes $< 30^\circ$ at a junction
(20° , 45° also tested)

- Gap: distance along the stroke between the two parks' attachment intervals (0 if they overlap)
- 9 settings: stroke angle $\in \{20, 30, 45\}^\circ \times \text{lim} \in \{400, 800, 1600\} \text{ m}$; E0 pairs excluded; network C
- Main street = stroke in the top 5% / 10% of network length by mean NACH-1600, $\text{NACH}_r = \log(\text{choice}_r + 1) / \log(\text{depth}_r + 3)$ — a cluster attribute, not an edge condition
- Groups: same procedure as all definitions, split to a 10-min walking diameter

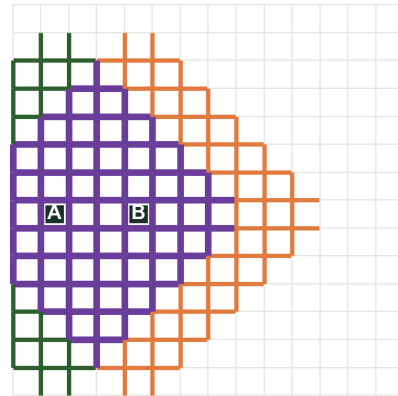
N3 · Formal definition

Near pair (3 blocks): $J_{ang} = 0.22$, ratio = 0.41 → no edge: proximity explains the overlap

few-turn fields: $J = 0.22$

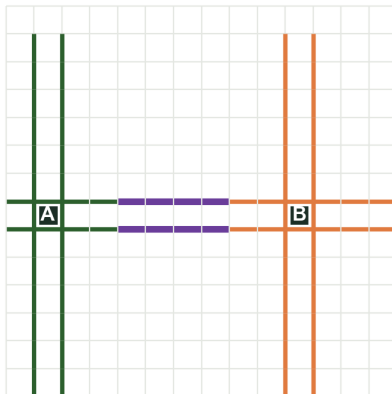


metric fields: $J = 0.52$

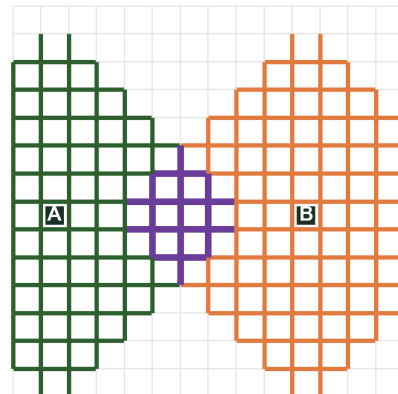


Far pair, same street (9 blocks): $J_{ang} = 0.10$, ratio = 1.25 → N3 edge ($\tau = 0.1$, $\kappa = 1.0$)

few-turn fields: $J = 0.10$



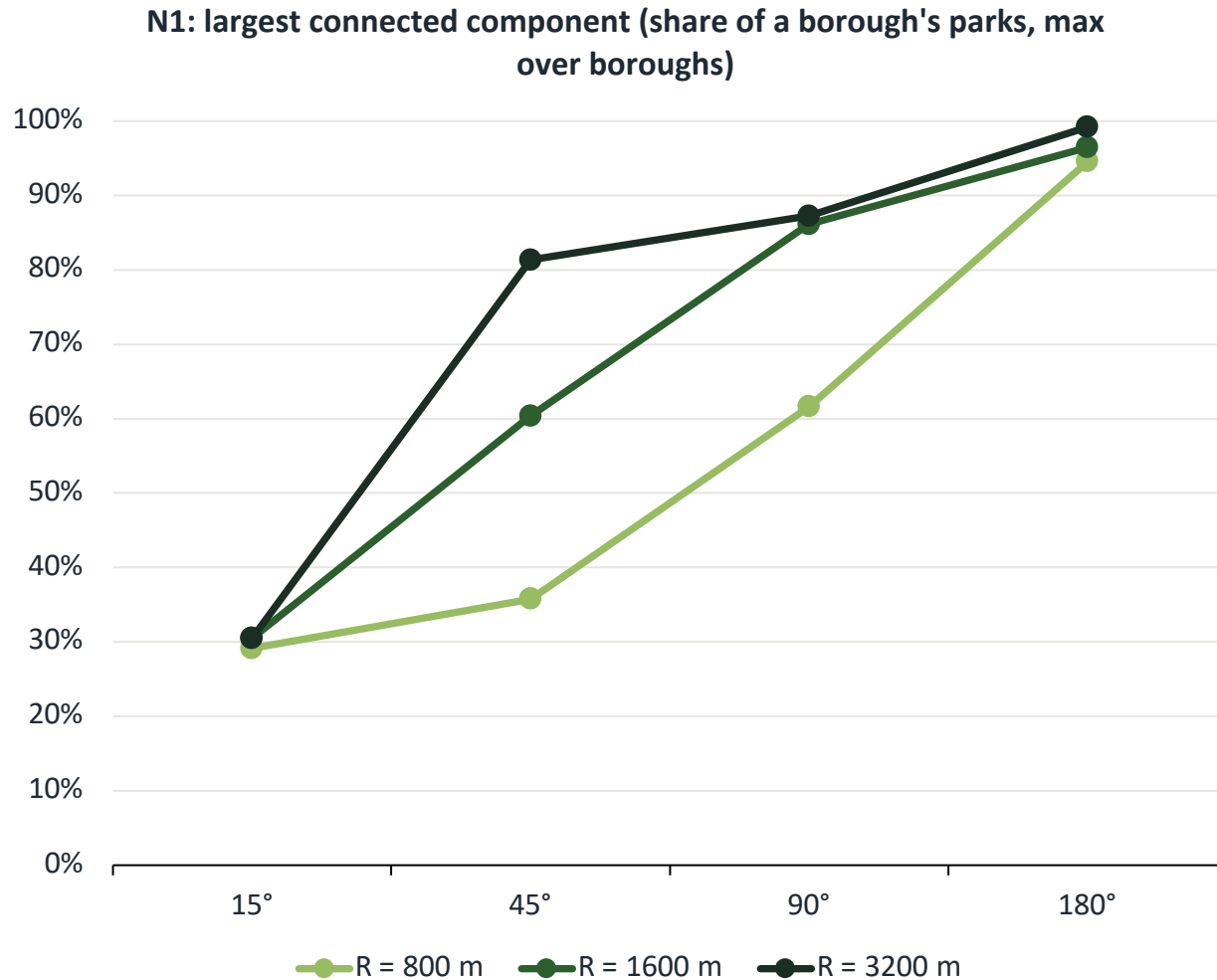
metric fields: $J = 0.08$



- Field = segments reachable with angular cost $\leq \theta c$ and path length $\leq R$
- J = length-weighted Jaccard of two parks' fields, for angular and metric fields
- Edge if $J_{ang} \geq \tau$ and $J_{ang} / J_{met} \geq \kappa$
- 16 settings: $\theta c \in \{45, 90\}^\circ \times R \in \{800, 1600\} \text{ m} \times \tau \in \{0.1, 0.2\} \times \kappa \in \{1.0, 1.25\}$

Toy grid: the near pair shares fields only because it is close (ratio 0.41); the far pair on one street keeps sharing its few-turn field (ratio 1.25).

From park graphs to robust groups



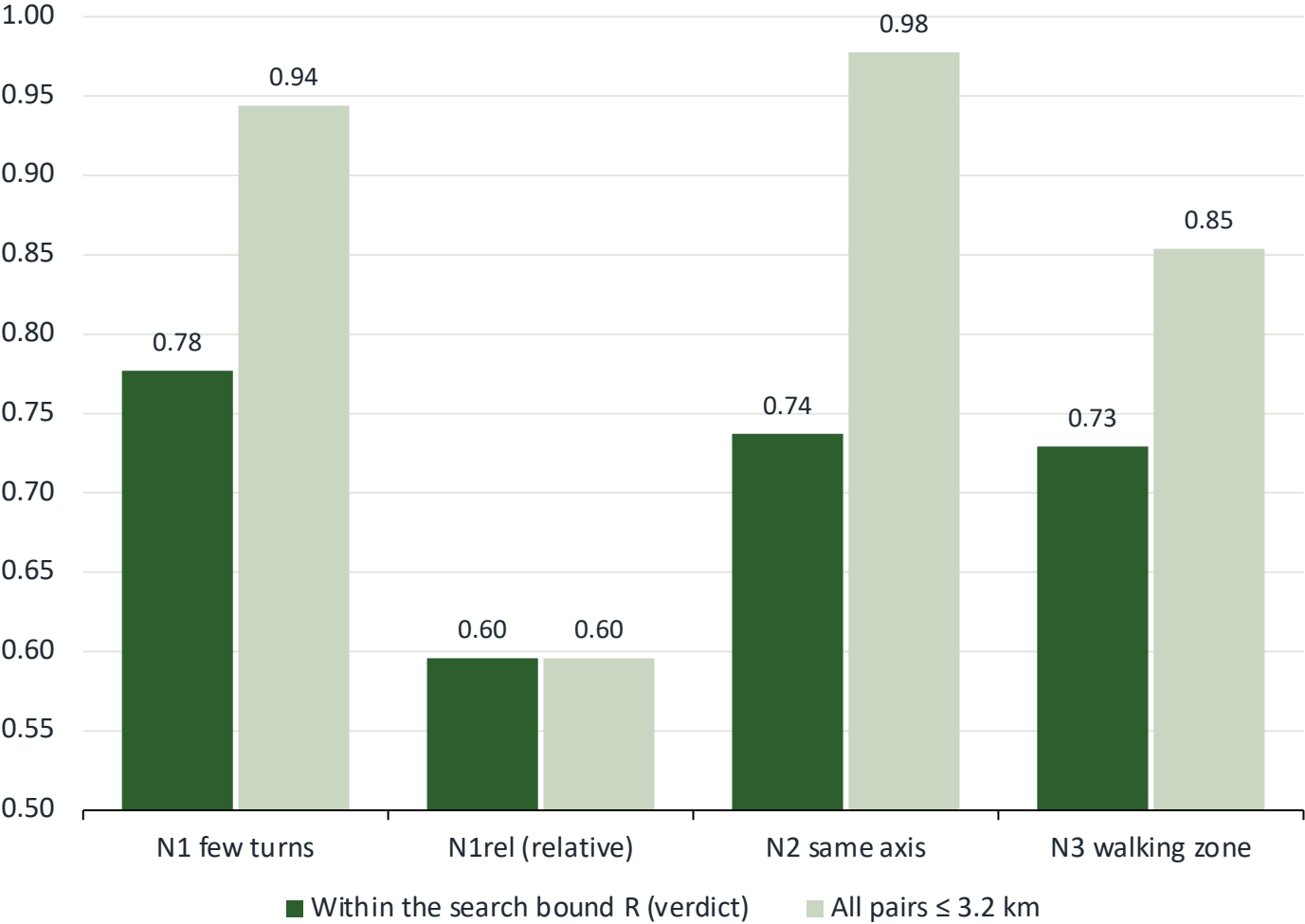
Percolation Once $\theta \geq 45^\circ$ at $R \geq 1.6$ km, or one turn is allowed, most of a borough's parks fall into one component; Queens and Staten Island percolate much later.

Communities In each setting, Louvain communities find internally dense sets of parks (resolution $\gamma = 1, 2, 4$).

Consensus Co-membership share across settings $\rightarrow$ average-linkage clusters (≥ 0.5) $\rightarrow$ split to a 10-min walking diameter.

Isolation IsolationPersistence = share of settings with no configurational edge; ≥ 0.8 = robust isolated.

Proximity check (AUC)



Proximity AUC: probability that a connected pair is closer than an unconnected one; > 0.85 = proximity-dominated.

Result: no definition crosses 0.85 once distance is held to its search bound.

Spearman (angular cost vs network distance) ≈ 0.55 .

Limitations

- Percolation: allowing one turn connects almost all parks in a borough — groups rely on strict settings and community detection
- Turn costs include bends inside segments (one turn $\approx 90\text{--}100^\circ$), so $\theta = 90^\circ$ is a strict one-turn threshold
- OSM completeness and mapping style (separately drawn sidewalks) affect angular costs; S results differ from C
- Park entrances unknown (boundary attachment); N2 computed on C only; long linear parks split into ~ 400 m pieces
- D: only the 200 m setting avoids percolation, so D groups are tight walking-distance groups
- N1 and N2 groups overlap substantially (ARI 0.69) — street-axis continuity is a stricter subset of few-turn links
- Groups not yet validated against observed park use or pedestrian counts